

OpenMDBench: Where Should the LLM Sit? A Distributed Open Benchmark for Layered LLM–Controller Architectures

Anonymous Author(s)

Submission Id: XXXX

Abstract

Hybrid architectures—large language models (LLMs) as deliberators, small models as executors—are increasingly adopted for *multi-domain* tasks, yet no existing benchmark can say *whether, when, and why* hybridization pays off. **OpenMDBench** answers with three contributions. (1) A semi-MDP formalization of the two-tier decision stack with an *observable performance-accounting framework*: exact identities decompose hybrid advantage against the strongest pure baseline into planning, execution, interaction, and baseline-gap terms. (2) A *distributed open evaluation platform* (central simulation, local inference) with two released instantiations: a semantically enhanced grid environment and a high-fidelity multi-domain adversarial suite admitted through *pre-registered acceptance gates*. (3) A *replay-gated counterfactual attribution method* that quantifies recoverable planning and execution losses and their non-additive interaction from a recorded episode.

Validation is two-sided. The grid pins the *negative boundary*: the goal interface is causally load-bearing, yet the deployed hybrids still trail the strongest pure baseline through a structural decision-form mismatch. The gate-designed adversarial suite confirms the *positive region*: layered stacks outperform every monolithic baseline on the majority of scenarios, end-to-end LLM control roughly halves the composite score, and a prompt-level audit shows deliberation-layer value hinges on information quality, not quantity.

Keywords

multi-agent systems, evaluation benchmark, large language models, hybrid LLM–small model architecture, distributed open evaluation, multi-domain heterogeneous systems

ACM Reference Format:

Anonymous Author(s). 2027. OpenMDBench: Where Should the LLM Sit? A Distributed Open Benchmark for Layered LLM–Controller Architectures. In *Proc. of the 26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027)*, Hanoi, Vietnam, 3–7 May 2027, IFAAMAS, 9 pages.

1 Introduction

Two problems, two decision layers. Multi-domain tasks pair a *deliberation* problem—whom to watch, what to intercept, how to respect interaction protocols, at second-to-minute cycles over semantic, open-ended states—with an *execution* problem—control,

tracking, sensing, and reactive avoidance, at millisecond-to-second cycles over continuous spaces. The boundary between the two decision layers coincides with the boundary between physical domains (aerial, surface, shore, underwater). Pure LLM agents cannot meet execution latency and cost budgets; pure RL agents cannot ingest open-ended language objectives or adversarial deception. **Hybrid LLM–small-model architectures—LLMs as deliberators, small models as executors—are the community’s convergent response** [5, 13, 14, 24].

Three questions, no ground to answer them. If hybridization is the future, benchmark-grade evidence for three questions is conspicuously absent:

- **(Q1) Payoff:** Does a hybrid stack actually outperform the best pure paradigm—and where does it *underperform*?
- **(Q2) Necessity:** Under what measurable task properties is layering *structurally necessary* rather than merely convenient?
- **(Q3) Bottleneck:** When a hybrid fails, is the fault in planning, in execution, or in the interface between them?

The obstacle is infrastructural: every existing multi-agent benchmark—from SMAC [19] to LLMarena [3]—inherits the closed, centralized submission model of the MARL era. For hybrid systems this fails computationally (centrally hosting billion-parameter LLMs is prohibitive), institutionally (proprietary weights cannot be uploaded), and architecturally (predefined interfaces constrain the composition design space hybrid research explores). Architectures are being built, but the community has no shared ground to compare them—and no instrumentation to tell *why* one wins.

OpenMDBench. We resolve the paradox with a requirements-driven pipeline—formalize (Section 3), build (Section 4), instrument (Section 5), validate (Section 6)—whose outputs map one-to-one onto three contributions. Validation runs on two instantiations sharing the full interface stack: a semantically enhanced grid environment maps the layering law’s boundary conditions (~1,000 episodes, complete planner×executor factorial), and a high-fidelity adversarial suite compares five stacks on 14 deception-enabled scenarios under a unified no-intelligence prompt regime. All reported numbers derive from released per-episode traces.

Contributions. Our contributions are threefold, mapping one-to-one onto the pipeline:

(C1) A formalization and two-sided test of the layering law (Section 3). The HSemi-MDP formulation, the three-form cross-domain coupling measure, and an observable performance-accounting framework whose deployment identities decompose hybrid advantage into planning, execution, and interaction terms against the strongest pure baseline—interface benefit stated as



This work is licensed under a Creative Commons Attribution International 4.0 License.

a falsifiable, task-dependent hypothesis. The grid instantiation *refutes the unconditional form* (“advantage scales with coupling”) and pins the law’s negative boundary: a goal interface that is causally necessary yet not sufficient for net advantage. The high-fidelity suite, designed via the same gates to sit in the positive region, *supports the conditional form*: layering outperforms every monolithic baseline on the majority of scenarios.

(C2) OpenMDBench, a distributed open evaluation platform with dual instantiations (Section 4). The polling-API architecture removing the three frictions above; an asymmetric four-domain coastal-protection scenario family (36 scenarios, 5 categories) with a layered, planning-vs-execution metric suite; and two released instantiations sharing the full interface stack—a semantically enhanced grid environment and a high-fidelity multi-domain adversarial suite (14 deception-enabled scenarios) admitted through pre-registered acceptance gates with version-frozen manifests. All artifacts are released with per-episode traces.

(C3) A diagnostic toolbox and an evidence-backed placement conclusion (Sections 5–6). A replay-gated counterfactual attribution method decomposing performance gaps into $\Delta P/\Delta E/\Delta I$ against matched reference components—validated on injected-bottleneck cases (5/5 graded faults; 20/20 anchored faults) and on a paired counterfactual of the real LLM goal stream (10/10 seeds)—plus a goal-dose interface-ablation protocol that operationalizes B_{if} causally. Across both instantiations the toolbox answers the placement question—*where should the LLM sit?* On the high-fidelity suite, layered stacks with mature executors beat every monolithic baseline on 11–12 of 14 scenarios (composite 0.77–0.78 vs. 0.63–0.72); end-to-end LLM control fails decisively (0.43 mean composite, per-scenario deficits up to 0.75); and a prompt-level audit traces a counterintuitive no-intelligence result to a single misleading directive—deliberation-layer value hinges on information quality, not quantity.

2 Related Work

MARL benchmarks. Classic cooperative MARL benchmarks [6, 19] standardize algorithm comparison but operate in single domains with homogeneous agents; heterogeneity efforts [9, 22, 26], application benchmarks [4, 18], social-dilemma platforms [12], and API/training standardizers [2, 10, 21] all remain single-domain and within the closed evaluation paradigm. None addresses hybrid LLM–small-model systems.

LLM multi-agent benchmarks. Interactive-LLM benchmarks [1, 3, 8, 11, 25, 27] (with theory-of-mind studies [20] alongside) evaluate LLM agents but run the LLM on the platform side—the inverse of our distributed paradigm—and none spans multiple physical domains or evaluates hybrid stacks.

Hybrid LLM–small model architectures. LAMARL [24], LGC-MARL [13], LERO [5], and reflective collaboration [14] demonstrate promising LLM-RL compositions—but provide no systematic evaluation framework for comparing designs, which is precisely what a benchmark must supply.

Distributed and federated evaluation. Distributed computation in RL has been training-side (distributed RL [17], IMPALA [7],

A3C [16], federated RL [23]); FEDAGENT [15] federates LLM-agent *training*. To our knowledge, no prior benchmark runs simulation centrally while agent inference runs locally—a new evaluation paradigm.

Comparison. Table 1 summarizes: no existing benchmark combines multi-domain heterogeneity, hybrid-architecture support, natural-language task interfaces, and an open evaluation paradigm.

3 Problem Definition: Decision-Stack Bifurcation and Formalization

3.1 The Two Layers

The cross-domain deliberation layer operates on tasks, intentions, and coordination: (i) mission understanding; (ii) cross-domain resource allocation; (iii) multi-platform scheduling; (iv) opponent intent inference; (v) rules-of-interaction compliance; (vi) dynamic re-planning. State spaces are semantic, objectives open-ended, cycles second-to-minute.

The intra-domain execution layer operates on platform control: attitude/heading/depth regulation, trajectory tracking with dynamic obstacles, sensor processing, reactive evasion and interference mitigation, domain-specific tactics. State and action spaces are continuous, rewards dense and well-defined, cycles millisecond-to-second.

The bifurcation is a property of the *problem*, not the solver (Section 1).

3.2 HSEMI-MDP Formalization

We formalize multi-domain adversarial systems as a **hierarchical semi-MDP** $M = \langle M_d, \{M_e^k\}, G \rangle$.

Upper layer: deliberation Semi-MDP. The tuple is $M_d = \langle S_d, A_d, P_d, R_d, \gamma_d, \tau_d \rangle$. The state $S_d = (X, Q, B, L)$ comprises the platform-task allocation matrix X , task-progress vector Q , opponent-intent belief B , and rules-of-interaction state L . Each action $a_d = \{(i, j, \theta)\}$ assigns platform i to task j with parameters θ ; actions are event-driven with random inter-decision intervals τ_d . Transitions P_d are induced stochastically by lower-layer outcomes; R_d encodes mission completion, rule-violation penalties, and coordination bonuses; γ_d matches second-to-minute cycles.

Lower layer: execution MDP family. For each domain $k \in \mathcal{D} = \{\text{aerial, surface, shore, underwater}\}$, $M_e^k = \langle S_e^k, A_e^k, P_e^k, R_e^k, \gamma_e^k \rangle$ with continuous physical states, domain-specific dynamics, and rewards $R_e^k(s, a; G_k, w_k)$ parameterized by the goal set G_k and priority weights w_k handed down from a_d ; γ_e^k matches millisecond-to-second cycles (10 Hz in our implementation).

Inter-layer interface—with explicit cost. The upper layer never emits low-level controls; it parameterizes lower MDPs via (G_k, w_k) , and lower-layer termination events (goal reached, platform loss, communication drop) trigger upper-layer updates. Unlike prior treatments that idealize this interface as free, we model a **carrying cost** $c \geq 0$: goal-conditioned execution is strictly weaker than an unconstrained monolithic policy when the goals carry no

Table 1: OpenMDBench vs. representative multi-agent benchmarks.

Benchmark	Domain	Agent type	Evaluation	LLM	Hybrid arch.	NL input
SMAC [19]	Single (StarCraft)	Homogeneous MARL	Closed centralized	✗	✗	✗
Melting Pot [12]	Single (grid)	Homogeneous MARL	Closed centralized	✗	✗	✗
PettingZoo [21]	Multiple	Mixed	Closed centralized	✗	✗	✗
LLMArena [3]	Single (games)	Pure LLM	Closed centralized	✓	✗	✓
BattleAgentBench [1]	Single (games)	Pure LLM	Closed centralized	✓	✗	✓
HIVE [11]	Single (RTS)	LLM + human	Closed centralized	✓	Partial	✓
OpenMDBench	Multi-domain (4)	Hybrid LLM–small model	Distributed open	✓	✓	✓

exploitable information (translation loss, latency, reduced reactivity). This single term, empirically identified below, reconciles why layering can hurt.

Adversarial extension—evaluated defender vs. scripted opponent. Red (defender—the *evaluated* side) and blue (opponent—a *standardized scripted agent*) run role-specific HSEMI-MDPs coupled through a shared environment state, with role-specific natural-language briefings. Fixing the opponent to a parameterized scripted adversary standardizes scenarios for cross-system comparability, halves evaluation cost, and realizes the D1 deception design, while its strength knobs double as the D2 headroom-calibration instrument; a community two-role extension remains supported but is not part of the evaluated protocol.

3.3 Cross-Domain Coupling and the Layering Law

A key structural parameter of any task instance is its **cross-domain coupling**—how much upper-layer decisions must coordinate across domains. Three complementary forms:

(1) *Action coupling.*

$$C_{\text{action}}(a_d) = |\{k \in \mathcal{D} : \exists i, \text{domain}(i) = k \wedge (i, \cdot) \in a_d\}|, \quad (1)$$

the number of domains touched by one allocation decision.

(2) *Intrinsic task coupling.*

$$C_{\text{intrinsic}}(\text{task}) = D_{\min}(\text{task}) \cdot I(\text{task}), \quad (2)$$

with $D_{\min}$ the minimum domain count required by task constraints and $I \in [0, 1]$ the inter-domain dependency strength (composition in App. A).

(3) *Effective coupling.*

$$C_{\text{effective}}(\pi, \text{task}) = \frac{1}{N} \sum_m C_{\text{action}}(a_d^m), \quad (3)$$

observable from execution traces; measures how much a policy *actually* coordinates.

Operationalizing the coupling terms. Two quantities anchor the law empirically: C_{info} , the information asymmetry between the deliberation layer’s observation channel and the execution layer’s, probed behaviorally through discrimination performance on the public observation channel (the D1 gate below), and B_{if} , the effective information carried by the goal interface—operationalized *causally* as the paired performance cost of disabling the interface (the goal-dose ablation of Section 5.2), not by an observational

point estimate. Both are grid-side instruments; the high-fidelity suite relies on the pre-registered gates.

The layering law. Let V_m be the performance of the best *pure-execution* (monolithic) baseline, and $\Delta V = V_{\text{hybrid}} - V_m$. Our experiments support the conditional form:

$$\Delta V = -c + f(C_{\text{info}}, B_{\text{if}}) \cdot \text{headroom}(V_m), \quad (4)$$

with $f \geq 0$ increasing in both arguments and $\text{headroom}(V_m) = (V^* - V_m)/(V^* - V_{\min})$ the normalized gap to the attainable ceiling V^* . Reading: layering pays only when (i) the interface carries genuine cross-layer information ($B_{\text{if}} > 0$ requires $C_{\text{info}} > 0$ asymmetric by construction), (ii) the interface’s carrying cost c is not yet saturated by that information, and (iii) the monolithic baseline leaves headroom. Prior unconditional formulations (“advantage scales positively with coupling”) omit c and headroom and are refuted in our boundary-regime experiments (Section 6.4): disabling the goal interface collapses paired performance—the interface is causally necessary—yet both deployed hybrids sit significantly *below* the strongest pure baseline, whose near-saturated score leaves little headroom.

Performance accounting. With V_{XY} the mean score of planner $X \in \{\text{rule}, \text{LLM}\}$ over executor $Y \in \{\text{rule}, \text{RL}\}$, define the matched contrasts $P = V_{\text{LR}} - V_{\text{RR}}$ (planner swap) and $E = V_{\text{RM}} - V_{\text{RR}}$ (executor swap), the non-additive remainder $I = V_{\text{LM}} - V_{\text{LR}} - V_{\text{RM}} + V_{\text{RR}}$, and the baseline gap $B = V_m - V_{\text{RR}}$. The deployed hybrids’ net advantages against the strongest pure baseline are then $\Delta V_A = V_{\text{LR}} - V_m = P - B$ and $\Delta V_B = V_{\text{LM}} - V_m = P + E + I - B$ —pure bookkeeping over the paired runs the benchmark already releases. The identities fix the threshold layering must clear; whether the interface itself pays remains a falsifiable, task-dependent hypothesis. Deployment accounting is distinct from the reference-component counterfactuals $\Delta P/\Delta E/\Delta I$ of Section 5.1.

Design corollaries (pre-registered acceptance gates). Like a pre-registered analysis plan, the law converts into falsifiable environment requirements, enforced before any full-scale run: **(D1)** information asymmetry (rule discriminator $\leq 60\%$, LLM $\geq 85\%$ on feint-vs-real); **(D1’)** decision-form match—core difficulty in LLM-strong decision forms, not spatial micro-control (greedy script $< 90\%$ of rule-baseline score); **(D2)** headroom (best pure baseline SR $\in [40\%, 85\%]$); **(D3)** interface budget (goal-dose ablation effects exceeding seed-level variance). Section 6.6 shows what happens when D1’ is violated; Section 6.7 embeds all four gates into the high-fidelity protocol.

4 The OpenMDBench Platform

4.1 Distributed Open Evaluation Architecture

The **Central Simulation Server (CSS)** maintains environments, generates natural-language mission briefings, executes actions, and computes metrics. **User-Local Hybrid Agents (ULHAs)** run all inference on user infrastructure, communicating through a standardized **polling API** (RESTful HTTP, JSON payloads; full specification in App. B). Figure 1 summarizes the architecture. Simulation advances asynchronously of inference, and sessions are resumable (verified under 120 s injected latency, App. B). This single decoupling removes all three frictions: no weights or code cross the boundary (institutional), users bear their own inference cost (computational), and the composition design space stays fully open (architectural). Integrity measures (rate limits, action-pattern consistency checks) bound the trust model, discussed in Section 7.2.

4.2 Scenario Family and Task Suite

We instantiate an *asymmetric coastal-protection* scenario: Red (the evaluated defender of Section 3.2) fields heterogeneous platforms with complementary sensing and endurance—UAVs (aerial), USVs (surface), shore radar, AUVs (underwater); Blue (the standardized scripted opponent) has concentrated capability with inferior sensing, deployed through tactic templates with calibrated strength knobs (App. C). The asymmetry forces intent inference under partial observability.

Task suite. 36 scenarios across 5 categories, each at 3 difficulties (108 configurations), spanning reconnaissance, tracking, interception, zone control, and emergency response. Every task is designed against three principles: cross-domain coordination is *required* (≥ 2 domains), intra-domain execution is *non-trivial*, and inter-layer coupling is *bidirectional*. Detailed parameters and per-category couplings: App. A, App. C, App. G.

Dual instantiations. The platform ships with two instantiations sharing the full interface stack (polling API, metric engine, two-role protocol). The **high-fidelity multi-domain adversarial suite** realizes the scenario family on a continuous-dynamics engine (three-layer scenario architecture, deception by construction, calibrated strength knobs); its 14 scenarios are admitted through the gates of Section 6.7. A **semantically enhanced grid-world port-protection** environment preserves the same protocol at minimal geometry for controlled boundary-regime experiments: deception is again by construction—feints indistinguishable in numeric features, resolvable only through repeated semantic contacts (App. H). Three difficulty tiers scale entity counts, action noise, communication degradation, and anomaly rates (full specification: App. H). Both instantiations are experimental substrates of Section 6.

4.3 Layered Evaluation Metrics

Single-scalar rankings cannot answer Q1–Q3. We compute a **layered suite** from ~ 40 raw per-episode measures (all dumped as traces for offline reanalysis): *planning-attributable* metrics (mission understanding, resource-allocation efficiency vs. the MILP

optimum, adaptability, cross-domain coordination, opponent-intent inference with feint detection); *execution-attributable* metrics (task success, time/energy efficiency, safety, tracking accuracy, friction statistics); and a *composite* score (default $\alpha/\beta = 0.5/0.5$), aggregated with paired statistics (Wilcoxon signed-rank, sign tests, bootstrap CIs; seeds paired across systems). Full definitions: App. E.

5 Diagnostic Instrumentation

Ranking cannot answer Q3. We instrument the benchmark with two causal tools that exploit the HSemi-MDP structure.

5.1 Replay-Gated Counterfactual Attribution

Method. For a system (π_d, π_e) with performance $V(\pi_d, \pi_e)$, we construct three counterfactuals: (1) *reference-execution replay*—record the upper decision sequence $D = \{(t_m, a_d^m)\}$, enforce those decisions at the same times, and replace execution with an independently validated reference executor π_e^{ref} ; (2) *reference planning*—replace π_d with a reference planner π_d^{ref} while keeping π_e (π_d^{ref} may adapt to execution feedback); (3) *reference full stack*. The method is *replay-gated*: a contrast is admitted only after the original episode and every counterfactual pass exact-replay verification on the deterministic engine, so each gap is attributable to the swapped component. The asymmetry between (1) and (2) follows from the time-scale separation: upper decisions are sparse and event-driven, so fixing their sequence does not over-constrain; lower actions are dense, so fixing them would erase the policy. The decomposition reads:

$$\Delta E = V(\pi_d, \pi_e) - V(\pi_d, \pi_e^{\text{ref}}) \quad (\text{execution}), \quad (5)$$

$$\Delta P = V(\pi_d, \pi_e) - V(\pi_d^{\text{ref}}, \pi_e) \quad (\text{planning}), \quad (6)$$

$$\Delta I = V(\pi_d^{\text{ref}}, \pi_e^{\text{ref}}) - V(\pi_d^{\text{ref}}, \pi_e) - V(\pi_d, \pi_e^{\text{ref}}) + V(\pi_d, \pi_e) \quad (\text{synergy}). \quad (7)$$

Validity guarantees. Five pre-registered checks (identical interfaces across counterfactuals; deterministic dynamics with fixed seeds; replay consistency; dose-response monotonicity; independently validated reference components) guard the decomposition; App. D states them in full. Section 6.3 reports the injected-bottleneck validation (graded faults recovered 5/5 with strict dose-response monotonicity) and an anchored-fault extension that localizes the layer of first appearance. The gate also holds on the live LLM planner: a matched full-hold intervention lowers the score in 10/10 paired seeds (Section 6.4).

Executor-effectiveness precheck. Attribution requires an executor that *responds* to goals; at a performance floor, planner differences are masked. We therefore gate attribution with an executor responsiveness test and report floor effects explicitly (App. D).

Determinism scope. Replay assumes a deterministic system, satisfied by the grid; the high-fidelity LLM endpoint is nondeterministic (Section 6.8), so there the diagnosis uses the prompt-level audit and stability criteria instead.

*Reserved for the architecture artwork:
central simulation server, user-local hybrid agents, and the polling API between them.*

Figure 1: Distributed open evaluation architecture: simulation and metrics run centrally; all agent inference runs locally behind a polling API, so no weights or agent code cross the trust boundary.

5.2 Goal-Dose Interface Ablation

B_{if} is operationalized causally. In the *goal-dose protocol*, each seed runs under paired interface conditions on the same executor and environment: *strong* (interface intact), *mask1/mask2* (one or two units’ goals per planning round replaced by a legal hold, masked units rotating by a fixed rule), and *hold* (every non-hold goal replaced by a legal hold). The primary estimand is the paired strong–hold contrast—the goal stream’s total contribution—with the masking tiers providing a unit-availability dose axis. Replaced commands are *legal* actions, so the intervention changes what the interface carries, not whether the executor can act; manipulation gates verify both goal acceptance and action change. Doses are unit-availability levels, not information bits: the protocol tests whether the interface is the pathway through which planning reaches execution; it neither estimates an information bandwidth nor fits f in Eq. (4).

6 Validation: Boundary Regime and Positive Region

We now validate the entire pipeline and pin down the layering law’s boundary regime. *All numbers derive from released per-episode traces; every table is paired by seed.*

6.1 Setup

Systems. Pure RL: MAPPO (CTDE, goal-conditioned; training details in App. F). Rule: scripted planner (confirmed-type filtering, nearest-interceptor assignment, sticky tracking). Pure LLM: the LLM emits low-level actions directly. Hybrid: LLM planner (JSON goal plans, replan every 10 steps, safe-mode fallback) $\times$ three executor backends (MAPPO / heuristic GOAI / reference). LLM backend: Qwen3.8-27B on vLLM (temperature 0.1; serving details in App. F). Protocols: 20 paired episodes per system (seeds paired), bootstrap 95% CIs, Wilcoxon/sign tests.

6.2 RQ1 & RQ3: Paradigm Payoff and the Bottleneck

Difficulty cliff—closed to end-to-end RL, open to orchestration. Across the three tiers, pure MAPPO scores 100% / $\approx 97\%$ / 0% (3 seeds each; complex exhaustively trained: 20M steps, curriculum learning, breach-and-draw shaping—all 0%), while the scripted rule planner scores 100% / 97% / 65%. The complex tier’s discriminative load (deception with indistinguishable numeric features,

anomaly relabeling, interference) is exactly where end-to-end small models collapse and orchestration survives.

The 2×2 factorial isolates the executor floor. On complex: every MAPPO-executor stack scores 0% (pure RL argmax 0%; rule+MAPPO 0%; LLM+MAPPO 0%, all losses by port breach); with the heuristic GOAI executor, rule reaches 65% while LLM reaches 5%. The MAPPO executor is the *common* bottleneck across all learning stacks—goal-conditioning cannot rescue an execution policy that never saw successful defense trajectories.

Tournament (round-robin, 3 difficulties $\times$ 5 episodes/pair, defender side). Defender Elo: pure RL 1555 $\approx$ hybrid 1553 (67% win rate: simple/medium carry, complex collapses) $>$ rule 1531 $\gg$ pure LLM 1331 $>$ random 1302; the scripted opponent rates 1728 (trace-role provenance in App. F). **On the grid instantiation, hybridization yields no net advantage** (-2.7 Elo)—consistent across four independent evidence chains. Per the layering law this is the expected *boundary* outcome, not a paradox: the grid tier violates D1’ (its difficulty is geometric interception, an LLM-weak decision form) and D2 (simple/medium saturated at $V_m \geq 0.95$, complex floored the RL executor).

6.3 Attribution Method: Injected-Bottleneck Validation

We corrupt a reference hybrid with graded, known faults and ask the method to label the bottleneck (Table 2).

Table 2: Injected-bottleneck validation of replay-gated counterfactual attribution: 5/5 correct labels; ΔE strictly monotone in injected severity (0.14 $\rightarrow$ 0.34 $\rightarrow$ 0.53)—validity checks (iii)/(iv) pass.

Injected fault	Expected	Recovered	Dose–response
Planner noise (goals)	planning	planning	$\Delta P = 0.54$
Executor degrad. $p=0.1$	execution	execution	$\Delta E = 0.14$
Executor degrad. $p=0.3$	execution	execution	$\Delta E = 0.34$
Executor degrad. $p=0.5$	execution	execution	$\Delta E = 0.53$
Balanced (both, mild)	balanced	balanced	$\Delta P \approx \Delta E = 0.07$

An *anchored-fault* extension sharpens the localization claim: each synthetic fault is anchored at a critical opportunity in a pre-matched trajectory prefix (state fingerprint matching the clean

run up to injection), with injection reach, local consequence, and endgame effect reported separately; the layer of first appearance is localized at the goal/action recording boundary in 20/20 paired seeds. These are known synthetic faults on a rule-planner base under matched prefixes—a calibration of the recording-and-replay pipeline, not blind natural-fault attribution.

6.4 RQ2: The Layering Law’s Boundary Regime

We probe the interface’s causal role on the medium tier (continuous authorization) with three paired batches, 10 seeds per condition, exploratory seed-bootstrap 95% CIs, contrasts strictly within batch, on an engine release and score metric kept separate from the Section 6.2 runs. *Goal-dose* (seeds 512–521) runs the rule- and LLM-planner hybrids over the same RL executor under the four conditions of Section 5.2; the *six-arm accounting* batch (seeds 501–510) crosses both planners with both executors, adds the two pure paradigms, and evaluates the identities of Section 3.3 (Table 3); the *real-stream counterfactual* (seeds 601–610) replays the live LLM planner’s goals against a matched full-hold intervention (Section 5.1).

Table 3: Six-arm accounting batch: mean scores, per-seed successes, and the deployment identities of Section 3.3 (seed-bootstrap 95% CIs). V_m = strongest pure baseline = rule+rule.

Stack	Mean V	Succ.	Identity [95% CI]
rule+rule ($V_{RR} = V_m$)	0.913	9/10	—
pure RL	0.883	9/10	—
rule+MAPPO (V_{RM})	0.810	8/10	$E = -0.103 [-0.263, 0.000]$
LLM+rule (V_{LR})	0.700	6/10	$P = -0.213 [-0.497, +0.080]$
LLM+MAPPO (V_{LM})	0.673	6/10	$I = +0.077 [-0.240, +0.380]$
pure LLM	0.280	1/10	—
$B = V_m - V_{RR} = 0; \Delta V_A = P - B = -0.213 [-0.497, -0.037];$			
$\Delta V_B = P + E + I - B = -0.240 [-0.490, -0.087].$			

The interface is causally necessary. Disabling it collapses paired performance: strong—hold = +0.457 for the rule-planner stack (95% CI [+0.247, +0.667]) and +0.537 for the LLM-planner stack ([+0.313, +0.737]); under hold both stacks score 0.213 (0/10 successes, 10/10 port breaches) and become *indistinguishable*—planner-source differences vanish exactly when the interface is cut. The real-stream counterfactual agrees: replacing the live LLM planner’s non-hold goals with holds lowers the score in 10/10 seeds (+0.420, CI [0.200, 0.637]); the matched rule planner still scores higher over the same executor (−0.260), so the stream’s contribution is distinct from planner quality.

Interface effectiveness is not net advantage. Table 3 evaluates the identities: against the strongest pure baseline ($V_m = 0.913$, 9/10), both ΔV_A and ΔV_B are significantly negative—each deployed hybrid sits *below* that baseline although its interface is causally load-bearing. This is the boundary regime of Eq. (4): a near-saturated baseline leaves little headroom, and the interface’s benefit does not clear its cost.

Doses refute the naive reading. Intermediate doses are non-monotone: light masking leaves the rule-planner stack *above* its unmasked score (0.833 vs. 0.670), and only 2/10 (rule) and 5/10 (LLM)

seeds are monotone across doses. “More interface information is always better” fails; the conditional form—a non-trivial f and a carrying cost c —remains the defensible reading.

6.5 Failure-Mode Anatomy: Three Interventions, One Structural Conclusion

If the LLM planner merely lacked information, injection should rescue it. We run a pre-registered intervention study on complex (20 paired episodes, four systems; Table 4).

Table 4: Intervention study on the complex tier (20 paired episodes). The informed LLM planner receives rendered heading kinematics and a six-point exclusive intelligence brief; performance does not move.

System	SR	Contrast	ΔSR [95% CI]
rule + heuristic	65%	—	—
rule_intel (feint filter)	60%	vs. rule	−5 [−15, 0]
LLM + heuristic	10%	—	—
LLM + exclusive intel	5%	vs. LLM	−5 [−20, +10]
		vs. rule	−60 [−85, −30]

The informed planner receives rendered heading kinematics and the six-point exclusive intelligence brief of Table 4 (feint behavior, own-goal rule, anomaly relabeling, interference guidance). Planner traces show the LLM *reads and cites* the intelligence—the information channel works—yet performance does not move (trace excerpts in App. H). Failure tracing locates the deficit in **semantic-to-geometric decision conversion**: recurring all-in commitments under spatial-combinatorial load, plus reliability friction (9% empty responses on max-token exhaustion). The frequency intervention (replanning 10 → 5 steps) and the interface intervention (Section 6.6) equally fail: all three channels—information, frequency, interface—are exhausted. **Conclusion (H1b evidence):** on geometrically shaped tasks the hybrid failure is structural; we formalize the missing property as **decision-form match (D1’)** and predict the high-fidelity suite’s combinatorial scheduling core sits on the LLM-favorable side.

6.6 Interface Modality: A Load-Matched Double-Edged Sword

Natural-language vs. JSON state rendering, paired seeds (full table in App. H): JSON wins at low semantic load (−16% NL–JSON gap at medium, $p = 0.98$); NL trends back at high load as JSON accrues parse failures. Interface choice is load-matched, not universal—and never rescues an overmatched planner (7–11% vs. rule’s 65% on complex).

6.7 From Boundary Conditions to the High-Fidelity Design

The grid results convert directly into the pre-registered acceptance gates of Section 3, calibrated on the grid—**D1** (rule discriminator $\leq 60\%$, LLM $\geq 85\%$), **D1’** (greedy-script $< 90\%$ of rule baseline), **D2** (best pure baseline SR $\in [40\%, 85\%]$), **D3** (goal-dose effects exceeding seed-level variance)—plus engineering gates (empty-response

and JSON parse-error $\leq 5\%$). Per-episode interface information is not re-estimated on the high-fidelity suite; the main run was preceded by per-scenario determinism gates, cross-arm channel-fairness audits, and manifest-hash freezing of engine and scenario versions.

6.8 High-Fidelity Adversarial Results: The Positive Region

The 14-scenario adversarial suite compares five stacks under a unified *no-intelligence* prompt regime: two layered stacks (LLM deliberation over a rule-based and an RL-based executor), end-to-end LLM control, and the two monolithic baselines (pure rule, pure RL). Each LLM stack runs five seeds; because the LLM endpoint is nondeterministic, a fixed seed is one *independent replication* rather than a paired replay (engine-side determinism is verified separately), so comparisons use a pre-registered three-part stability criterion—mean-win, every-seed-win, and split-half-win. The baselines never read the briefing channel and are reused byte-identical from the frozen archive (SHA-256 verified). Table 5 reports per-scenario composite scores; Figure 2 the end-to-end gap distribution. Three findings emerge.

Layering wins where the gates hold—with disclosed exceptions. Both layered stacks beat every monolithic baseline on the majority of scenarios (11–12 of 14; overall composite 0.77–0.78 vs. 0.63–0.72), and the full three-part criterion confirms 8 (LLM+Rule) and 9 (LLM+RL) of the 28 stack–baseline–scenario comparisons as stable. The exceptions are visible in Table 5 rather than averaged away: pure RL dominates the surface-raid scenario IE-03 (0.96, its best cell); the headroom term predicts the smallest gain there, App. A); the RL-executor stack is further handicapped by an ammunition-doctrine cap that the pure-RL baseline does not use, making that cell structurally incomparable. The rule baseline wins three scenarios against each layered stack (IE-03/04/08 vs. LLM+RL; IE-04/08/10 vs. LLM+Rule). Hybrid advantage is the modal outcome, not a universal one.

End-to-end LLM control fails decisively and stably. Pure LLM averages 0.43 composite—roughly half the layered stacks’ 0.77–0.78—and trails the rule-executor layered stack on 14/14 scenarios (median deficit 0.33, worst 0.75; Figure 2). The deficit is structural and non-tunable, confirming the decision-form diagnosis of Section 6.5 from the negative side: the same LLM that plans winning allocations cannot itself convert them into track, intercept, and fire control.

Removing prompt intelligence does not hurt—and slightly helps (Table 6): stripping all opponent intelligence from the prompts leaves the layered stacks intact while the end-to-end stack *loses* 0.09—declared intelligence is a crutch only when no execution layer exploits it. A prompt-level audit traces the asymmetry to a single misleading directive in the legacy template: a substring heuristic labeled one scenario’s armed main-assault platforms (whose behavior text merely mentions decoys) as declared non-threats. Deliberation-layer value is governed by information *quality*, not quantity—and it is the audit machinery, not aggregate scores, that localizes the cause.

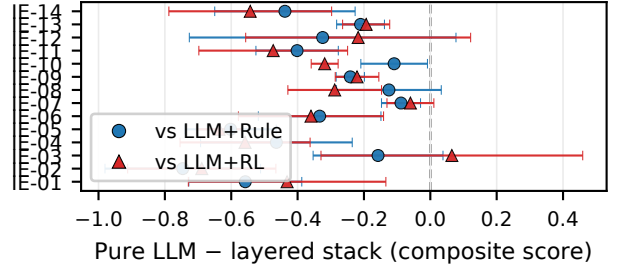


Figure 2: End-to-end LLM control deficit: per-scenario composite-score gap of Pure LLM against each layered stack (mean $\pm$ sd over five seeds; negative = layered stack ahead).

Table 6: Intelligence-regime ablation on the three LLM stacks (same 14 scenarios): the no-intelligence regime vs. the legacy regime declaring wave-by-wave opponent intelligence (including future ground truth); LLM+RL legacy episodes use the same locked checkpoint; n = episodes per regime; positive Δ = no-intelligence scores higher.

Stack	No-intel	Legacy-intel	Δ	Better on	n
LLM+Rule	0.774	0.753	+0.021	9/14	70/82
LLM+RL	0.783	0.782	+0.000	8/14	70/49
Pure LLM	0.431	0.516	-0.085	4/14	70/40

7 Discussion

7.1 Insights

I1 — Layering is conditional, and the conditions are testable. Hybrid advantage is neither universal nor mysterious: it requires information the execution layer cannot see, an interface cheap enough to carry it, and headroom in the monolithic baseline. The grid instantiation sits at the law’s boundary—a causally necessary interface that still does not pay at a near-saturated baseline; the high-fidelity suite, admitted through the same gates, lands inside the positive region—falsifiably and by design.

I2 — The bottleneck is executor-shaped, and single metrics hide it. The 2×2 factorial shows one weak executor floors every planner stacked above it, while the same planners under a heuristic executor separate cleanly (Section 6.2). End-to-end rankings would conflate the two; attribution plus factorial design disentangles them.

I3 — LLM failure on the grid is a decision-form mismatch, not an information deficit. All three rescue channels (exclusive intelligence, faster replanning, modality switch) fail; planner traces show the intelligence is read and cited (App. H). LLMs cannot yet convert semantic understanding into spatial-combinatorial scheduling under adversarial timing—environments seeking to credit LLM deliberation must place task difficulty in LLM-strong decision forms.

I4 — Benchmarks should ship their falsification criteria. Pre-registered hypotheses, paired statistics, released traces, and manipulation-validity checks made our own theory’s refutation

Table 5: High-fidelity adversarial suite (14 scenarios, unified no-intelligence prompt regime): composite defender score (normalized weighted average over applicable metric layers; higher is better). Bold: best stack per scenario. Each LLM stack runs five seeds (7/11/13/17/19; one episode per scenario–seed), a fixed seed one independent replication under the nondeterministic endpoint. Rule and RL never read the briefing and are reused from the frozen archive (SHA-256 verified; 3–42 episodes per scenario); LLM+RL uses a single locked checkpoint. Overall: mean $\pm$ sd across the 14 scenario means (equal scenario weighting).

Scenario	LLM+Rule	LLM+RL	Pure LLM	Rule	RL
IE-01 single target	0.900	0.774	0.343	0.691	0.876
IE-02 dual threat	0.908	0.851	0.162	0.765	0.757
IE-03 surface raid	0.856	0.634	0.699	0.748	0.960
IE-04 combined arms	0.710	0.804	0.246	0.812	0.687
IE-05 multi-axis	0.790	0.824	0.188	0.763	0.703
IE-06 decoy mixed	0.767	0.793	0.434	0.657	0.652
IE-07 cross-domain	0.792	0.763	0.703	0.749	0.684
IE-08 island strike	0.404	0.568	0.280	0.432	0.301
IE-09 staggered waves	0.880	0.859	0.638	0.820	0.629
IE-10 dual-axis pincer	0.712	0.921	0.603	0.801	0.729
IE-11 decoy screen	0.747	0.820	0.346	0.681	0.184
IE-12 fog onset	0.766	0.659	0.441	0.711	0.641
IE-13 deep strike	0.807	0.790	0.597	0.709	0.538
IE-14 saturation	0.793	0.897	0.354	0.787	0.506
Overall	0.774 $\pm$ 0.124	0.783 $\pm$ 0.100	0.431 $\pm$ 0.187	0.723 $\pm$ 0.098	0.632 $\pm$ 0.203
Mean wins vs. Rule	11	11	0	–	–
Mean wins vs. RL	12	12	4	–	–

productive; we propose this as standard practice for evaluation-benchmark papers.

7.2 Limitations and Threats to Validity

Instantiation and estimator scope. The grid is deliberately minimal-geometry; its boundary-regime results do not estimate the positive region. The high-fidelity main results cover the 14-scenario interception-engagement family—one of the five designed categories—with five stacks under one no-intelligence prompt regime: a subset of the 36-scenario family and the full system matrix. The admitted scenarios span aerial, surface, and shore platforms; the engine implements all four domains including underwater AUVs, and remaining categories, two-role play, underwater scenarios, and community baselines proceed under the same admission gates. On the grid, goal doses are unit-availability levels, not bits; we do not fit f in Eq. (4), and all grid batches report within-batch paired contrasts only (Section 6.4). An earlier observational mutual-information estimator for B_{if} was retired after a seed-level audit (clipped negative estimates; between-seed variance exceeding its contrasts); the causal ablation protocol replaces it.

Backend and physics fidelity. All LLM results use one 27B open model (deployment reproducibility); planner-side generality is untested, though the rule-planner replication isolates architecture-level from model-level effects. Because the LLM endpoint is nondeterministic, fixed seeds are independent replications rather than paired replays; stability rests on the multi-seed criterion of Section 6.8. Physics fidelity is engine-level (continuous dynamics with tactic templates), not aerodynamic or hydrodynamic certification.

Trust model. Distributed evaluation cannot verify claimed architectures; rate limiting and action-pattern consistency checks bound but do not eliminate exploitation—the openness/control trade-off is deliberate.

8 Conclusion

Driven by multi-domain adversarial requirements rather than architectural fashion, OpenMDBench formalizes decision-stack bifurcation (HSEMI-MDP), pairs it with an auditable performance-accounting decomposition, and distills the conditional layering law. Its platform removes the frictions excluding hybrid LLM–small-model systems from existing benchmarks; its toolbox turns evaluation from ranking into causal analysis. Validation is two-sided: the grid maps the law’s negative boundary—a causally necessary yet insufficient interface, and LLM planners failing structurally on geometrically shaped tasks—while the high-fidelity suite, admitted through gates derived from those findings, confirms the positive region: layering beats every monolithic baseline on the majority of scenarios, end-to-end LLM control roughly halves the composite score, and deliberation-layer value hinges on information quality, not quantity. We release all environments, traces, and analysis code.

References

- [1] Anonymous. 2024. BattleAgentBench. *arXiv preprint* (2024).
- [2] Marc Bou, Pierluca D’Oro, Soumya Kar, et al. 2023. BenchMARL: A Benchmark for Multi-Agent Reinforcement Learning. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*.
- [3] Junzhe Chen, Yuming Liu, Xin Chen, et al. 2024. LLMarena: Evaluating Large Language Models in Dynamic Multi-Agent Environments. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*.

- [4] Marco Dominici, Lorenzo Bisi, Lorenzo Sabbatini, et al. 2025. HIVEX. In *International Conference on Learning Representations (ICLR)*.
- [5] Yutong Duan, Zhi Liu, Qian Zhang, et al. 2024. LERO. *arXiv preprint* (2024).
- [6] Benjamin Ellis, Mikayel Samvelyan, Robert Faulkner, et al. 2023. SMACv2: An Improved Benchmark for Cooperative Multi-Agent Reinforcement Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- [7] Lasse Espeholt, Hubert Soyer, Rémi Munos, et al. 2018. IMPALA: Scalable Distributed Deep-RL with Importance Weighted Actor-Learner Architectures. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*.
- [8] Yao Fu, Hao Peng, Tong Zhang, et al. 2025. Decrypto. In *International Conference on Learning Representations (ICLR)*.
- [9] Jun He, Yu Wang, Zhi Liu, et al. 2024. HeMAC: A Heterogeneous Multi-Agent Challenge Benchmark. *arXiv preprint* (2024).
- [10] Siyi Hu, Yifan Zhong, Quan Min, et al. 2022. MARLLib: A Scalable and Efficient Multi-Agent Reinforcement Learning Library. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*.
- [11] Yifei Hu, Wenyuan Li, Zhaoyang Zhang, et al. 2024. HIVE. *arXiv preprint* (2024).
- [12] Joel Z. Leibo, Edgar Dueñez-Guzmán, John P. Agapiou, et al. 2021. Melting Pot 2.0. *arXiv preprint arXiv:2107.06857* (2021).
- [13] Shuo Li, Han Wang, Jing Zhao, et al. 2024. LGC-MARL. *arXiv preprint* (2024).
- [14] Yixin Liang, Chen Song, Xin Liu, et al. 2024. Reflective Multi-Agent Collaboration with Large Language Models. *arXiv preprint* (2024).
- [15] Yang Liu, Shuai Wang, Jie Zhang, et al. 2024. FEDAGENT. In *AAAI Workshop on Federated Learning*.
- [16] Volodymyr Mnih, Adrià Puigdomènech Badia, Mehdi Mirza, et al. 2016. Asynchronous Methods for Deep Reinforcement Learning. In *Proceedings of the 33rd International Conference on Machine Learning (ICML)*.
- [17] Arun Nair, Praveen Srinivasan, Sam Blackwell, et al. 2015. Massively Parallel Methods for Deep Reinforcement Learning. *arXiv preprint arXiv:1507.04296* (2015).
- [18] Thanh Nguyen, Ruizhe Guo, Andrew Plumtpe, et al. 2023. APE: An Anti-Poaching Environment for Multi-Agent Reinforcement Learning. In *Proceedings of the 22nd International Conference on Autonomous Agents and MultiAgent Systems (AAMAS)*.
- [19] Mikayel Samvelyan, Tabish Rashid, Christian Schroeder de Witt, Gregory Farquhar, Nantas Nardelli, Tim G. J. Rudner, Chia-Man Hung, Philip H. S. Torr, Jakob Foerster, and Shimon Whiteson. 2019. The StarCraft Multi-Agent Challenge. In *Proceedings of the 18th International Conference on Autonomous Agents and MultiAgent Systems (AAMAS)*.
- [20] Natalie Shapira, Ateret Azran, Gabriel Stanovsky, et al. 2023. Theory of Mind for Multi-Agent Collaboration. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*.
- [21] Justin Terry, Benjamin Black, Nathaniel Grammel, et al. 2021. PettingZoo: Gym for Multi-Agent Reinforcement Learning. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- [22] Hao Wang, Jun Zhang, Shuai Li, et al. 2024. POAC. *arXiv preprint* (2024).
- [23] Xiaofei Wang, Zehui Xiong, Chenhan Xu, et al. 2023. Federated Reinforcement Learning: A Survey. *IEEE Transactions on Neural Networks and Learning Systems* (2023).
- [24] Xin Wang, Yu Zhang, Long Chen, et al. 2024. LAMARL. *arXiv preprint* (2024).
- [25] Wei Zhang, Xin Liu, Yang Li, et al. 2024. SPIN-Bench. *arXiv preprint* (2024).
- [26] Xiao Zhang, Ming Li, Yu Chen, et al. 2024. LSHC: A Large-Scale Heterogeneous Multi-Agent Cooperation Benchmark. *arXiv preprint* (2024).
- [27] Yifan Zhou, Jiannan Wu, Ziqiao Lin, et al. 2024. MindAgent. In *International Conference on Learning Representations (ICLR)*.